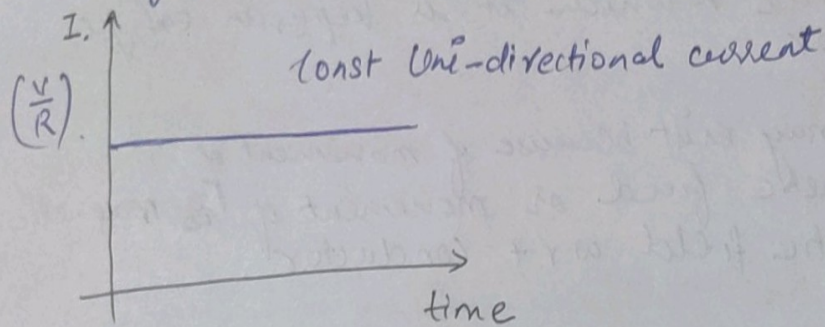
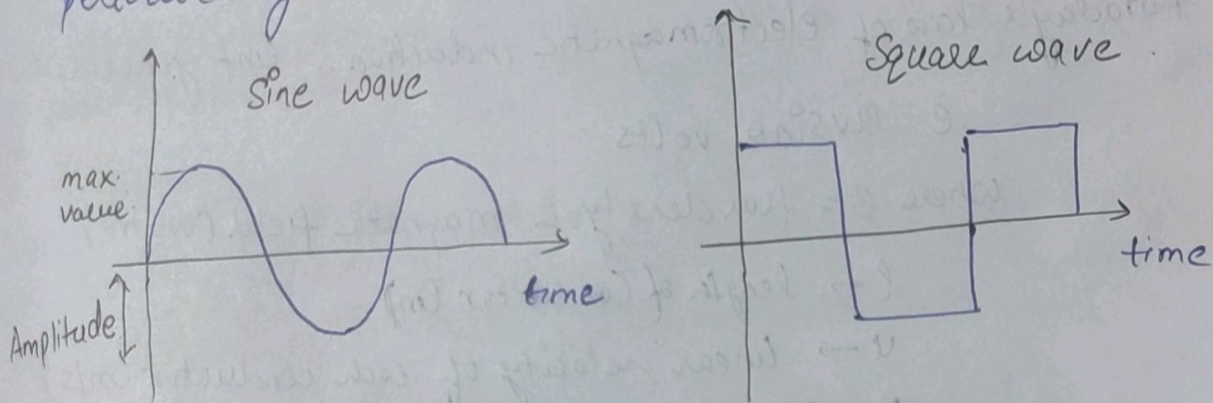


ALTERNATING CURRENT

* Direct Current (D.C) - It is a current which remain constant in magnitude w.r.t time.



* Alternating Current (A.C) - is a current which changes periodically w.r.t time both in magnitude and dirⁿ.



* Advantages of A.C.

- ① Voltage in a.c system can be raised or lower with help of transformer. In d.c system, raising and lowering of voltages is not so easy.
- ② Electrical transmission of a.c signals at high voltage is possible.
- ③ A.c electrical motors are simple in construction, are cheaper & require less maintenance.
- ④ whenever it is necessary a.c supply can be easily converted to obtain d.c supply.

Generation of A.C voltage.

• Faraday's law of electromagnetic Induction

It says that whenever there is relative motion b/w conductor and the magnetic field in which it is kept, an emf get induced in conductor.

The relative motion may exist because of movement of conductors w.r.t magnetic field or movement of ~~the~~ magnetic field or movement of magnetic field w.r.t conductor.

- If conductor motion of conductor is $\perp$ to dirn of magnetic field then we will get maximum flux and maximum voltage.

Accto Faraday's law of electromagnetic induction, emf generated is

$$e = Blv \sin \theta \text{ volts.}$$

where B = flux density of magnetic field (wb/m^2)

$l \rightarrow$ length of Conductor (m)

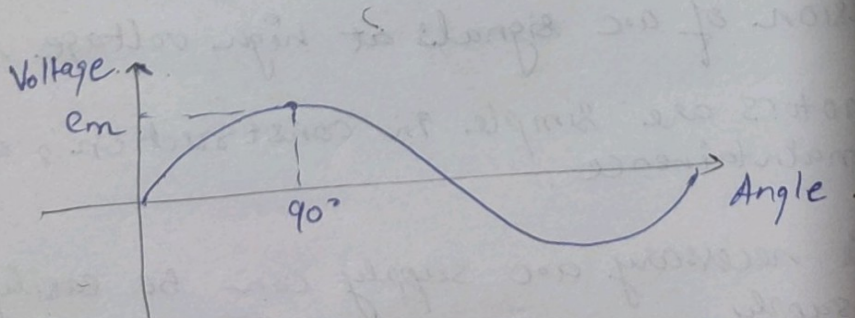
$v \rightarrow$ linear velocity of each conductor (m/s).

$E_m = Blv = \text{max value of induced emf.}$

* Note:-

$v \cos \theta$ is component which is parallel to magnetic field. due to which No flux generated and emf is zero.

$$\underline{\text{emf}} \leftarrow \boxed{e = E_m \sin \theta}$$



* Standard Terminology

1. Instantaneous Value:-

The value of an alternating quantity at particular instant is known as instantaneous value.

2. Amplitude:-

The maximum value attained by alternating quantity during positive or negative half cycle is called its amplitude.

3. Cycle:-

Each repetition of set of positive and negative instantaneous values of alternating quantity is called cycle.

4. Time Period:-

The time taken by an alternating quantity to complete its one cycle is known as its time period denoted by T seconds.

5. Frequency:-

The no of cycles completed by an alternating quantity per second is known as its frequency. It is denoted by f and. It is measured in cycles/seconds or Hertz (Hz).

The relationship b/w frequency and time period is

$$f = \frac{1}{T}$$

6. Angular Frequency:-

It is frequency expressed in electrical radians per second.

$$\omega = 2\pi f \text{ radians/sec.}$$

$$\text{or} \\ \omega = \frac{2\pi}{T} \text{ radian/sec.}$$

* Single phase AC:-

→ It is basic connection b/w source & load through a wire is s.

* Generation of single phase AC.

→ It is based on principle of Faraday's law.

→ When there is relative motion b/w magnet & conductor then emf is induced.

and emf, $e = -N \frac{d\phi}{dt}$ → magnetic flux.

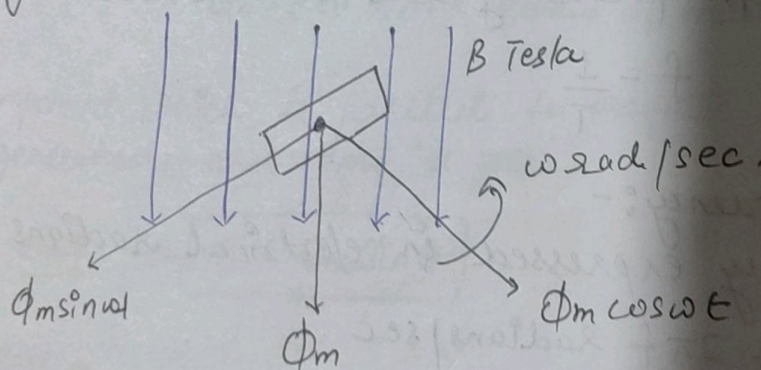
$\frac{d\phi}{dt}$ → Rate of flow of $\vec{B}$ through the conductor
No. of turns = constant

* EMF eqn of single phase AC.

$$e = -N \frac{d\phi}{dt}$$

• For deriving the formula for instantaneous emf,

Let us assume that we are having uniform magnetic field and having flux density B Tesla.



• Maximum flux linking with conductor when $\perp$ to $\vec{B}$ is ϕ_m .

$$e = -N \frac{d\phi}{dt}$$

For given condⁿ,

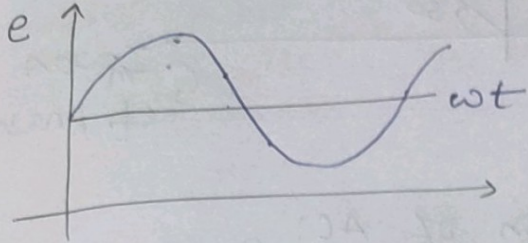
$$\phi = \phi_m \cos \omega t \quad [\text{Ir component}]$$

$$\frac{d\phi}{dt} = -\phi_m \sin \omega t \times \omega$$

$$e = -N \frac{d\phi}{dt} = -N \cdot -\phi_m \omega \cdot \sin \omega t$$

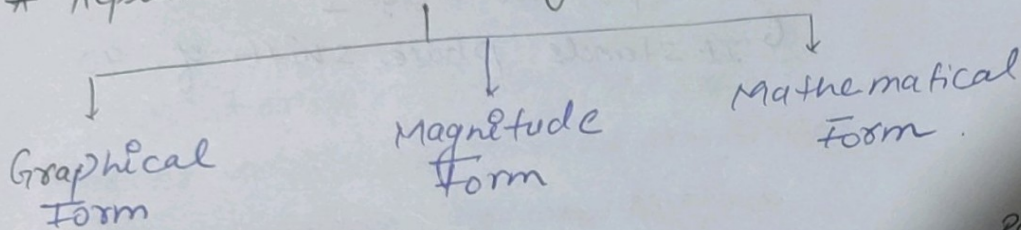
$$e = (N\omega\phi_m) \sin \omega t \Rightarrow \boxed{e = E_m \sin \omega t} \quad \{ E_m = N\omega\phi_m \}$$

$\downarrow$
 maximum flux.
 $\downarrow$
 Instantaneous emf



Induced voltage, $V = V_m \sin \omega t$ — (1)
 " current, $I = I_m \sin \omega t$ — (2) } Instantaneous Value.

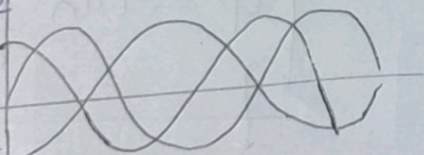
* Representation of Single phase AC



1. Graphical Form — (a) waveform $\rightarrow$ connection of various point traces in two axis.

Condⁿ:- to trace more than 1 waveform

$\rightarrow$ frequency of all quantities should be same.

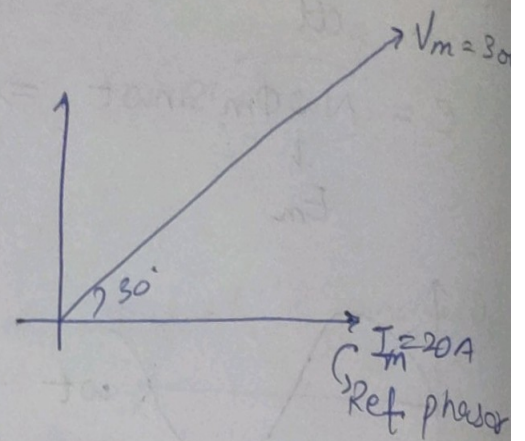
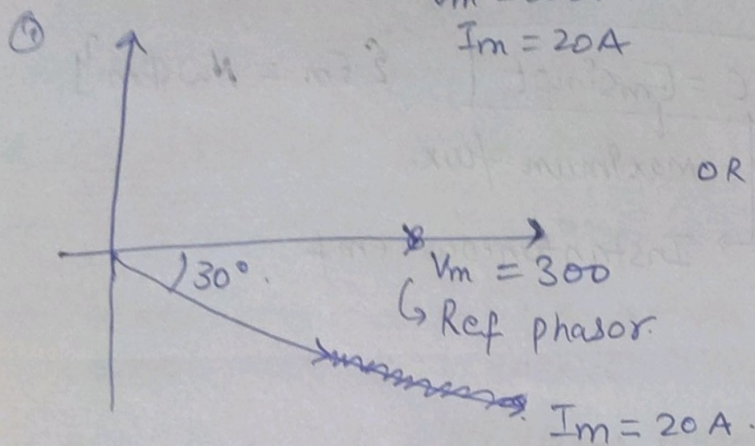


$\{$ x-axis $\rightarrow$ phase angle, time duration
 y-axis $\rightarrow$ magnitude of alternating quantity.

(b) Phasor:-

Assumption — we take ref phasor on x-axis
 $\rightarrow$ A line of definite length is drawn which represent the max. value of quantity.

Condⁿ reqd:- (1) We need max value of quantity
 (2) Need phase angle of all quantity
 (3) Need to know who is lagging/leading.



② Mathematical form of representation of AC: Reactance X_L or X_C

→ (a) Rectangular form:-

$\rightarrow$ Leading condⁿ
 $\rightarrow$ Lagging condⁿ

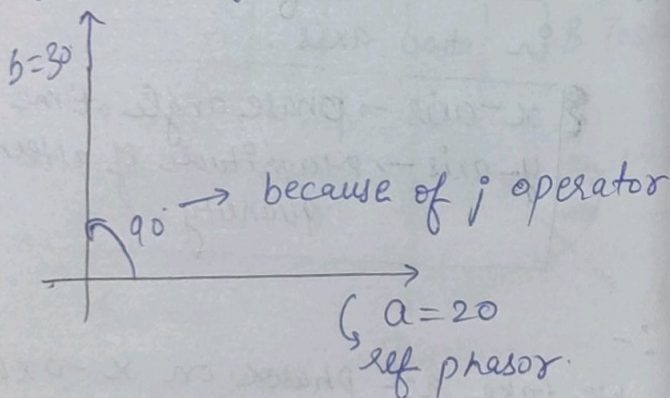
$a \pm jb \rightarrow$ complex no.
 $\rightarrow$ constant part or fluctuating part of alternating or Resistance quantity

$j \rightarrow$ operator $\equiv \sqrt{-1}$

$\rightarrow$ It stands phase shift of 90°

Ex $a = 20$
 $b = 30$

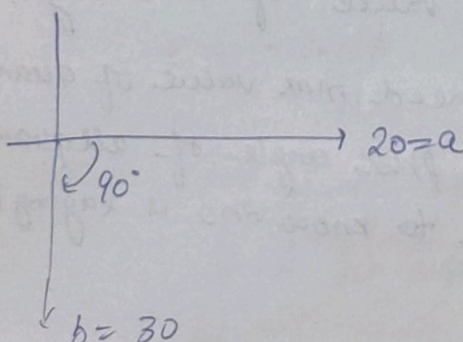
$a + jb = 20 + j30 \rightarrow$ expressed in phasor.



$$\phi = \tan^{-1} \frac{b}{a}$$

↓
phase angle

Ex $20 - j30$



(b) Trigonometric form:-

• Potential can be expressed as

$$V = a \cos \theta + j b \sin \theta$$

(c) Polar Form:-

$$V = |V| \angle \pm \theta$$

Ex. $V = |V| \angle +\theta$

→ V is leading I by θ

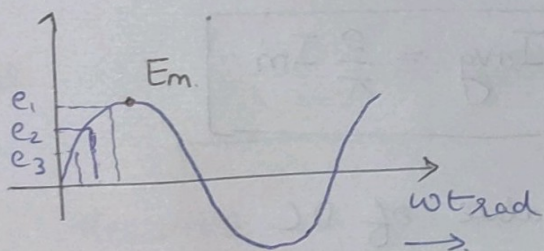
Ex. $V = |V| \angle -\theta$

→ V is lagging I by θ

(3) Magnitude Form:-

1. Instantaneous Velocity / Value
2. Peak or maximum value / Amplitude.
3. Average Value representation.
4. RMS Value.

(a) Instantaneous Velocity / value



$$\left. \begin{aligned} e &= E_m \sin \omega t \\ v &= V_m \sin \omega t \\ i &= I_m \sin \omega t \end{aligned} \right\} \text{Instant Value.}$$

→ Instantaneous value can be expressed in its peak value & angle in sine f^n

Ex. $v = \underbrace{200}_{V_m} \sin \underbrace{35t}_{\omega} \text{ V}$

$v = V_m \sin \omega t$

(b) Peak Value $\rightarrow (E_0, E_m, E)$.

Emf Peak Value $\rightarrow E_0, E_m, E$

Voltage Peak Value $\rightarrow V_0, V_m, V$

Current " " $\rightarrow I_0, I_m, I$

(c). Average value representation.

$\rightarrow$ It is sum of all instantaneous value at all instant divided by no of instant value.

$\rightarrow$ We find the avg value for half cycle.

Average Value:-

Taking area of alternation of $0-\pi$ and dividing it into all possible divisions:-

$$I_{avg} = \frac{\int_0^{\pi} \frac{I_m \sin \omega t}{\pi - 0} \cdot d\omega t}{\pi - 0} = \int_0^{\pi} \frac{I_m}{\pi} \sin \omega t \cdot d\omega t$$

$$I_{avg} = \frac{I_m}{\pi} \int_0^{\pi} \sin \omega t \cdot d\omega t = \frac{I_m}{\pi} [\cos \omega t]_0^{\pi}$$

$$= \frac{I_m}{\pi} \left[-\cos \frac{2\pi}{T} \times \frac{T}{2} + \cos 0 \right]$$

$$= \frac{I_m}{\pi} [2] = \frac{2I_m}{\pi} \quad \boxed{I_{avg} = \frac{2}{\pi} I_m}$$

1. RMS value of AC $\rightarrow$ Equivalent value of DC.

$$i = I_m \sin \omega t$$

Squaring both sides

$$i^2 = I_m^2 \sin^2 \omega t$$

$$i^2 = I_m^2 \left(\frac{1 - \cos 2\omega t}{2} \right)$$

Taking mean of given value.

$$i_{avg} = \frac{1}{2\pi} \int_0^\pi I_m (1 - \cos 2\omega t) d\omega t$$

$$i_{avg}^2 = \frac{I_m^2}{2} \int_0^\pi \frac{(1 - \cos 2\omega t) d\omega t}{\pi - 0}$$

$$i_{avg}^2 = \frac{I_m^2}{2\pi} \left(\left[\omega t \right]_0^\pi - \left[\frac{\sin 2\omega t}{2} \right]_0^\pi \right)$$

$$= \frac{I_m^2}{2\pi} (\pi - 0) = \frac{I_m^2}{2}$$

$$i_{avg}^2 = \frac{I_m^2}{2}$$

→ Square Rooting both sides

$$i_{rms} = \frac{I_m}{\sqrt{2}}$$

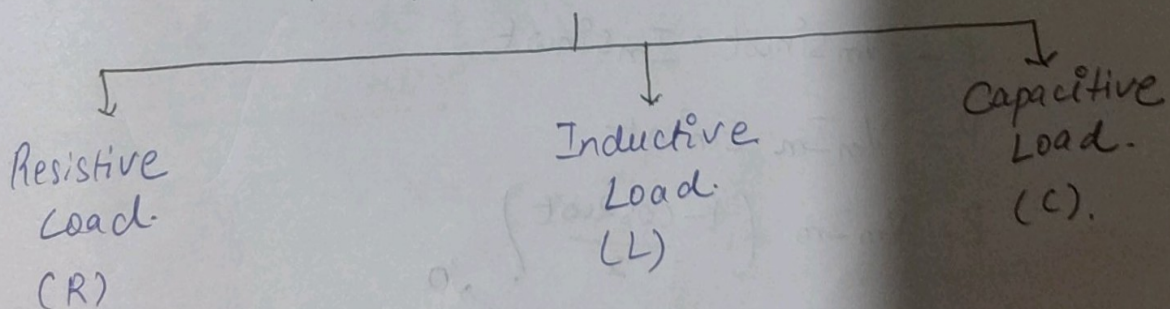
*Note:-

In ques, by default, value alternating quantity is given in its rms value.

Similarly, $V_{rms} = \frac{V_m}{\sqrt{2}}$

#

ANALYSIS OF SINGLE PHASE AC



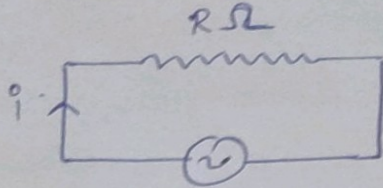
• We will study 2 things about electric component

(a). Relationship b/w v & I due to that load

(b). Power dissipated.

(a) AC circuit containing Purely resistive Load:-

①



$$V = V_m \sin \omega t \text{ V}$$

Let us assume that a purely resistive load $R\Omega$ is connected across a supply of

$$V = V_m \sin \omega t \quad \text{--- (1)}$$

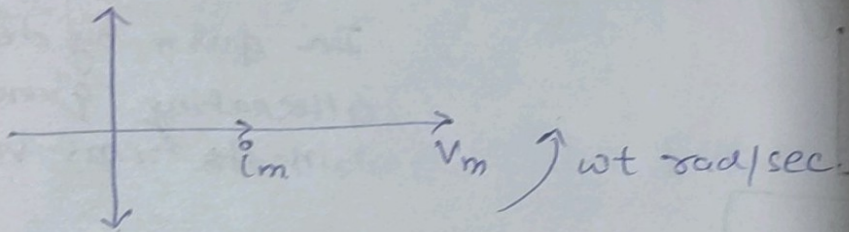
By Ohm's Law,

$$I = \frac{V}{R} = \frac{V_m \sin \omega t}{R}$$

$$i = \frac{V_m}{R} \sin \omega t \Rightarrow \boxed{i = I_m \sin \omega t} \quad \text{--- (2)}$$

On comparing ① & ②, we can say that voltage & current are in same phase.

• Phasor



② Power dissipated in purely resistive load

$$P = Vi$$

$$P = V_m \sin \omega t \cdot I_m \sin \omega t$$

$$P = V_m I_m \sin^2 \omega t$$

$$P = V_m I_m \left(\frac{1 - \cos 2\omega t}{2} \right)$$

$$P = \frac{V_m I_m}{2} - \frac{V_m I_m \cos 2\omega t}{2}$$

Neglecting fluctuating part

$$\boxed{P = \frac{V_m I_m}{2}}$$

$$P = V_m I_m \int_0^{2\pi} \frac{1 - \cos \omega t}{2\pi} \cdot d\omega t$$

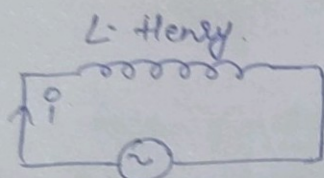
$$P_{avg} = \frac{V_m I_m}{2\pi} [\pi - 0] = \frac{V_m I_m}{2}$$

$$P_{avg} = \frac{V_m I_m}{2}$$

$$\text{so } P = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \Rightarrow \boxed{P = V_{rms} \cdot I_{rms}}$$

(b) AC circuit containing purely Inductive Load.

①



$$V = V_m \cdot \sin \omega t$$

Unit of Inductance is Henry.

Let us assume a purely inductive Load to be connected across a supply of

$$V = V_m \cdot \sin \omega t \quad \text{--- ①}$$

EMF induced in coil is given as

$$\boxed{e = -N \cdot \frac{d\phi}{dt}}$$

$$e = \frac{-V}{1} \quad \cdot \quad N = L \quad , \quad \frac{d\phi}{dt} = \frac{di}{dt}$$

$$-V = -L \cdot \frac{di}{dt} \Rightarrow V = L \cdot \frac{di}{dt}$$

$$V_m \sin \omega t = L \frac{di}{dt} \Rightarrow \int \frac{V_m \sin \omega t}{L} \cdot dt = \int di$$

$$i = \frac{V_m}{L} \left[-\frac{\cos \omega t}{\omega} \right] = -\frac{V_m}{\omega L} \cos \omega t$$

$$i = -\frac{V_m}{\omega L} \cos \omega t$$

$$i = -\frac{V_m}{\omega L} \sin(90 - \omega t)$$

$$i = \frac{V_m}{\omega L} \sin(\omega t - 90)$$

Comparing it with ohm's law

$$X_L = \omega L \quad [\text{By ohm's law}]$$

$$i = \frac{V_m}{X_L} \sin(\omega t - 90)$$

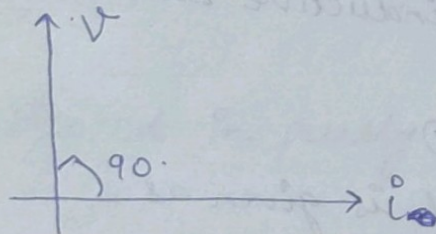
$X_L \rightarrow$ Reactance

$$i = I_m \sin(\omega t - 90^\circ) \quad \text{--- (2)}$$

Comparing (1) & (2), we get

i is lagging V by 90° or i lag from V by 90

* Phasor



(2) Power in AC circuit with purely inductive load

At

$$P = v i$$

$$P = V_m \sin \omega t \cdot I_m \sin(\omega t - 90^\circ)$$

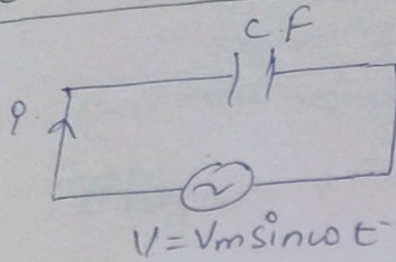
$$P = \frac{V_m I_m}{2} \sin \omega t \cos \omega t$$

$$P = \frac{V_m I_m}{2} \sin 2\omega t$$

neglecting fluctuating part

$$P = 0$$

(c) AC circuit with purely Capacitive Load:-



Let us assume that purely inductive load of 'C' farad is connected across a supply of

$$V = V_m \sin \omega t \quad \text{--- (1)}$$

The charge stored in capacitor is given as

$$q = CV$$

$$i = \frac{dq}{dt} = \frac{d(CV)}{dt}$$

$$i = C \cdot \frac{dV}{dt} = C \cdot \frac{d(V_m \sin \omega t)}{dt}$$

$$i = V_m C \cdot \cos \omega t \times \omega$$

$$i = V_m C \omega \cos \omega t$$

$$i = \omega C V_m \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$i = \frac{V_m \sin \left(\omega t + \frac{\pi}{2} \right)}{\frac{1}{\omega C}}$$

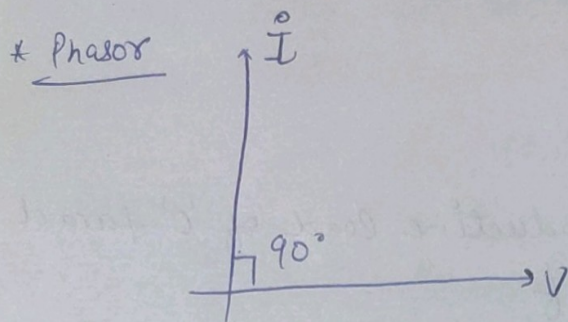
Now comparing it with ohm's law.

$$\boxed{X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}}$$

$$i = V_m X_c \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$i = I_m \sin \left(\omega t + \frac{\pi}{2} \right) \quad \text{--- (2)}$$

Comparing ① & ②, we get
 $\rightarrow i$ leads from V by 90°



* Power dissipated in Capacitive Load

$$P = VI \cos \phi$$

$$P = V_m \sin \omega t \cdot I_m \cos \omega t$$

$$P = \frac{V_m I_m}{2} \sin 2\omega t$$

~~Q.2~~ Neglecting fluctuating part

$$\boxed{P = 0}$$

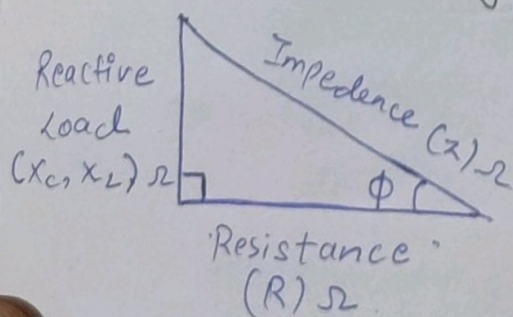
Components of Single phase A.C.

1. Impedance Triangle

2. Power Triangle

3. Power Factor

(a) Impedance Triangle



$$R : \phi = 0$$

$$X_L : \phi = 90^\circ$$

$$X_C : \phi = 90^\circ$$

$$Z = \sqrt{R^2 + X^2} \quad \{ X = X_L \text{ or } X_C \}$$

$$\textcircled{1} Z = \sqrt{R^2 + (X_C - X_L)^2} \quad ; \quad X_C > X_L$$

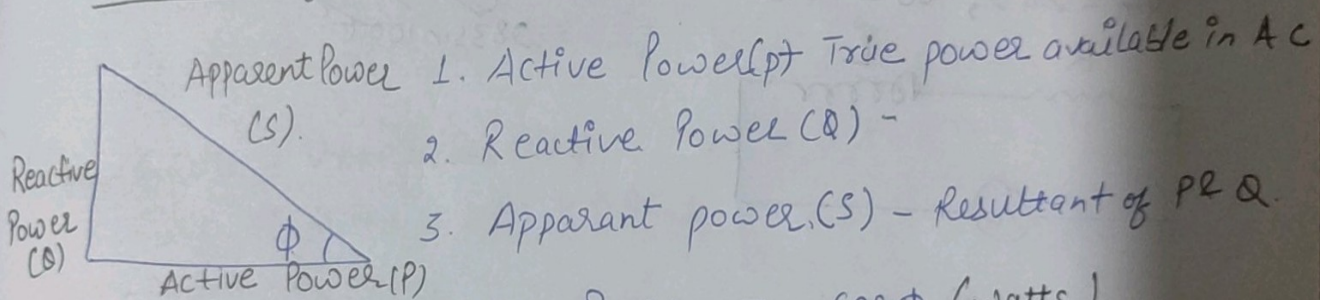
$$\textcircled{2} Z = \sqrt{R^2 + (X_L - X_C)^2} \quad ; \quad X_L < X_C$$

$$\textcircled{3} Z = R \quad ; \quad X_L = X_C \text{ (Condition for Resonance)}$$

$$\sin \phi = \frac{X}{Z} \quad \{ X = X_L \text{ or } X_C \}$$

$$\cos \phi = \frac{R}{Z}$$

(b) Power Triangle:-



Due to resistance $\leftarrow \textcircled{1} P = V_{rms} I_{rms} \cos \phi \text{ (watts)}$

Due to reactance / reactive component $\leftarrow \textcircled{2} Q = V_{rms} I_{rms} \sin \phi \text{ (Volt-ampere Reactive (VAR))}$

Due to overall impedance $\leftarrow \textcircled{3} S = V_{rms} I_{rms} \text{ (volt-ampere) (VA)}$

* Note:-

• If type of power is not mentioned then we assume Active Power.

(c) Power Factor:-

It is defined as the cosine of phase difference angle b/w supply voltage and current.

$$PF = \cos \phi$$

① $\cos\phi$ & ϕ is given

② $\cos\phi = \frac{R}{Z}$

③ $\cos\phi = \frac{\text{Active power}}{\text{Apparent power}}$

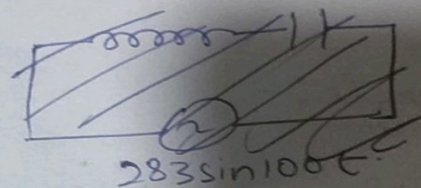
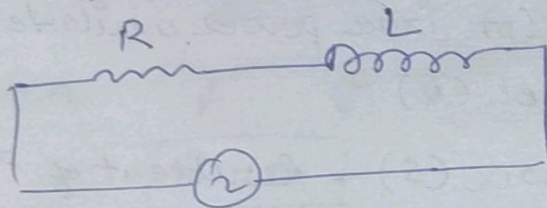
ways to find Power factor.

Numerical on Series AC Circuits: -

Q1. Find an expression for current flowing and calculate the power in series R-L circuit of value $R=50\Omega$ and $L=0.159H$ when voltage of $V=283\sin 100t$ is applied.

Ans $V=283\sin 100t$

$V_m = 283, \omega = 100$



$V = V_m \sin \omega t$

$i = I_m \sin \omega t + I_m \sin(\omega t - 90^\circ)$
 $i = I_m [\sin \omega t + \sin(\omega t - 90^\circ)]$

$X_L = \omega L$

$X_L = 100 \times 0.159$

$X_L = 15.9, R = 50\Omega$

$Z = \sqrt{(2500)^2 + 256}$

$Z = \sqrt{2756}$

$Z = 52.49\Omega$

$i = I_m \sin \omega t$

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{283}{\sqrt{2}} = 200.1V$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{200.1}{52.4}$$

$$I_{rms} = 3.81 A$$

$$I_m = I_{rms} \times \sqrt{2}$$

$$I_m = 5.39 A$$

$$\cos \phi = \frac{R}{Z} = \frac{50}{52.4} \Rightarrow \phi = 17.63^\circ$$

In R-L circuit, ϕ due to R is 0°

and ϕ due to L is 17.63°

In case of L, i lag v by ϕ .

$$I = I_m \sin(\omega t - 17.63^\circ)$$

$$i = 5.39 \sin(\omega t - 17.63^\circ)$$

$$\begin{aligned} \text{② Power} &= V_{rms} I_{rms} \cos \phi \\ &= 200.1 \times 3.81 \times 0.953 \end{aligned}$$

$$P = 726.5 \text{ watt}$$

Q2. If a resistor of 15Ω and Capacitor of $150 \mu\text{fd}$ are connected in series across 230V , 50Hz supply then Calculate

- ① Impedance
- ② Current
- ③ Power factor
- ④ Phase angle
- ⑤ Power consumed

Ans $R = 15 \Omega$, $C = 150 \mu\text{fd}$.

$V_{rms} = 230\text{V}$, $f = 50\text{Hz}$.

$$\omega = 2\pi f = 100\pi$$

$$1). Z = \sqrt{R^2 + X_C^2}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 150 \times 10^{-6}}$$

$$X_C = 21.2 \Omega$$

$$Z = \sqrt{225 + 450.7}$$

$$Z = 26 \Omega$$

$$\textcircled{2} \text{ Also } V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

$$V_m = 230 \times 1.414$$

$$V_m = 325.22$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z_T} = \frac{230}{26} = 8.85 \text{ A}$$

$$\boxed{I_{\text{rms}} = 8.85 \text{ A}}$$

$$\textcircled{3} \cos \phi = \frac{R}{Z} = \frac{15}{26} \quad \text{2d FL}$$

$$\cos \phi = 0.576$$

$$\textcircled{4} \phi = 54.83^\circ$$

$$\textcircled{5} P = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$= 230 \times 8.85 \times 0.576$$

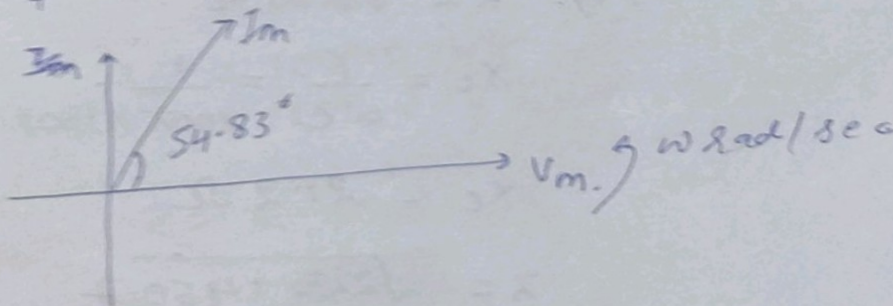
$$\boxed{P = 1172.45 \text{ watt}}$$

Phasor

$$V_m = 325.22$$

$$I_m = I_{\text{rms}} \times \sqrt{2} = 12.515 \text{ A}$$

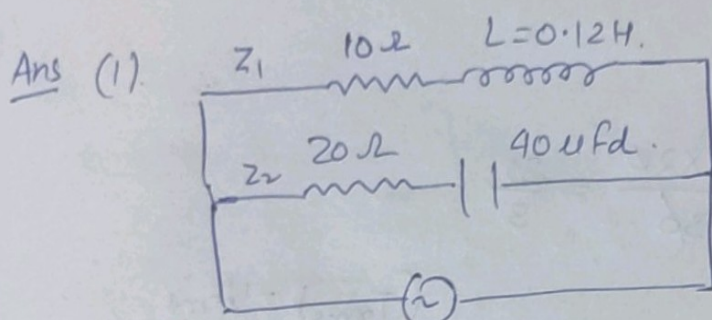
$$\phi = 54.83^\circ$$



* NUMERICALS ON PARALLEL AC CIRCUIT.

Q1 Two branches A & B are connected ^{in parallel.} across a supply of 200V, 50Hz. If A consist of resistance 10Ω & inductance of $0.12H$; B consist of resistance of 20Ω and Capacitor of $40\mu F$. then Calculate

1. Total Impedence
2. Total Current
3. Current in each branch
4. Phasor
5. Power factor



200V, 50Hz

$$V_{rms} = 200$$

$$f = 50Hz$$

$$\omega = 100\pi$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 40 \times 10^{-6}} = 79.6$$

$$X_L = \omega L = 100\pi \times 0.12$$

$$X_L = 37.68$$

$$\textcircled{1} Z = \sqrt{R^2 + X^2}$$

$$Z_1 = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{100 + (37.68)^2}$$

$$\boxed{Z_1 = 39.002\Omega}$$

$$Z_2 = \sqrt{R^2 + X_C^2}$$

$$Z_2 = \sqrt{400 + (79.6)^2}$$

$$\boxed{Z_2 = 82.051\Omega}$$

$$Z_T = \frac{Z_1 \times Z_2}{Z_1 + Z_2} \quad \{\text{for parallel AC circuit}\}$$

$$\nrightarrow Z_T = \sqrt{R^2 + (X_L - X_C)^2} \quad \{\text{for series}\}$$

$$(2) I_{rms} = \frac{V_{rms}}{Z_T} = \frac{200}{26.435}$$

$$I_{rms} = 7.565 A$$

(3) Current in branch 1

$$\text{Branch A: } (I_{rms})_1 = \frac{V_{rms}}{Z_1} = 5.127 A$$

$$\text{Branch B: } (I_{rms})_2 = \frac{V_{rms}}{Z_2} = 2.437 A$$

(4) Power factor

$$\cos \phi = \frac{R_T}{Z_T}$$

$$R_T = \frac{10 \times 20}{36} = \frac{20}{3}$$

$$\cos \phi = \frac{20}{3 \times 26.4} = 0.252$$

$$\cos \phi = 0.252$$

$$\phi = \cos^{-1}(0.252)$$

$$\phi = 75.4^\circ$$

$$(I_{rms})_1 = \frac{V_{rms}}{Z_1}$$

$$(I_{rms})_1 = \frac{200}{39.002} = 5.127$$

$$(I_{rms})_2 = \frac{200}{82.051} = 2.437$$

$$(I_{rms})_1 \times \sqrt{2} = (I_m)_1$$

$$(I_m)_1 = 7.24 A$$

$$(I_m)_2 = 2.437 \times \sqrt{2}$$

$$(I_m)_2 = 3.436 A$$

$$I_m = 7.565 \times \sqrt{2}$$

$$I_m = 10.69 A$$

(5) Phasors

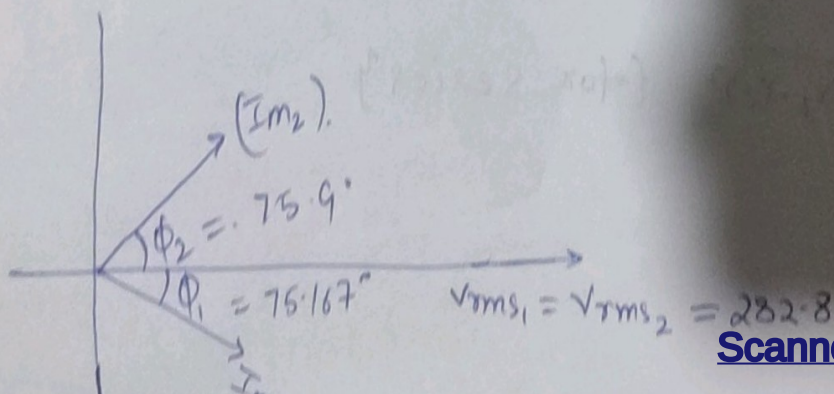
$$V_{rms1} = 200$$

$$I_{rms} = 7.565$$

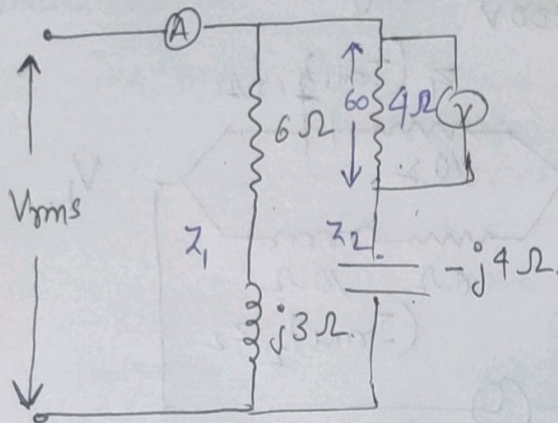
$$V_m = 200 \times \sqrt{2}$$

$$V_m = 282.8 V = V_{m2}$$

Phasor



Q3 In the given circuits, if a voltmeter reads 60V, then find reading of ammeter



(I) Ans $(I_{rms})_T = ?$

$$Z_1 = \sqrt{36 + 9}$$

$$Z_1 = 3\sqrt{5} \Omega$$

$$Z_2 = \sqrt{16 + 16}$$

$$Z_2 = 4\sqrt{2} \Omega$$

$$(I_{rms})_1 = \frac{V_1}{Z_1}$$

$$I_{4\Omega} = \frac{60}{4} = 15A$$

$$(I_{rms})_2 = 15A = \frac{(V_{rms})_2}{Z_2}$$

$$(V_{rms})_2 = 15 \times 4\sqrt{2}$$

$$= 60 \times 1.414$$

$$(V_{rms})_2 = 84.84V$$

$$(I_{rms})_1 = \frac{(V_{rms})_1}{Z_1} = \frac{84.84}{3\sqrt{5}} = \frac{28.28}{\sqrt{5}}$$

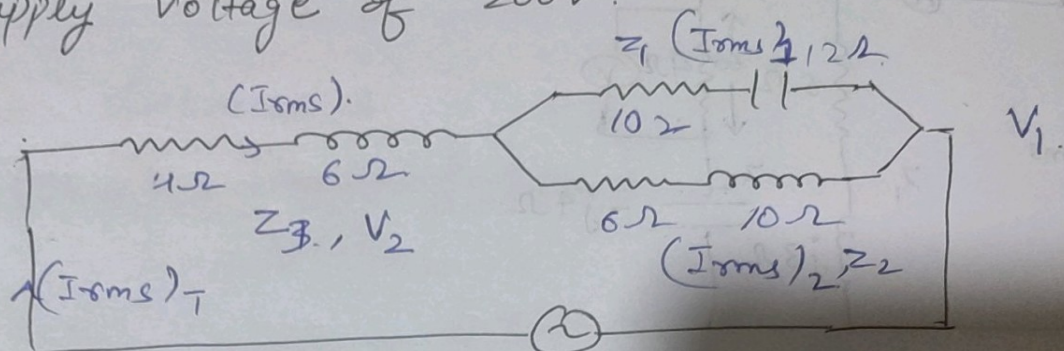
$$(I_{rms})_1 = \frac{28.28}{2.23} = 12.68$$

$$(I_{rms})_T = 12.68 + 15$$

$$(I_{rms})_T = 27.68A$$

* Numericals on Series & parallel & combined AC circuit

Q1 Determine the current drawn by given network at a supply voltage of 200V.



Ans

$$X_L = 12\Omega$$

$$X_C = 10\Omega$$

$$V_{rms} = 200V$$

$$\omega = 100\pi$$

$$V_{rms} = 200V, 50Hz$$

$$\underline{Z_1} = \sqrt{100 + 144}$$

$$= \sqrt{244} = 15.6\Omega$$

$$Z_2 = \sqrt{36 + 100}$$

$$= \sqrt{136} = 11.6\Omega$$

$$Z_3 = \sqrt{16 + 36}$$

$$= \sqrt{54} = 7.34\Omega$$

$$Z_T = Z_3 + \frac{Z_1 \times Z_2}{Z_1 + Z_2}$$

$$Z_T = 7.34 + \frac{15.6 \times 11.6}{27.2}$$

$$(V_{rms})_1 = (I_{rms})_1 \times Z_1$$

$$(I_{rms})_T = \frac{(V_{rms})_T}{Z_T}$$

$$= \frac{200}{14} = \frac{100}{7}$$

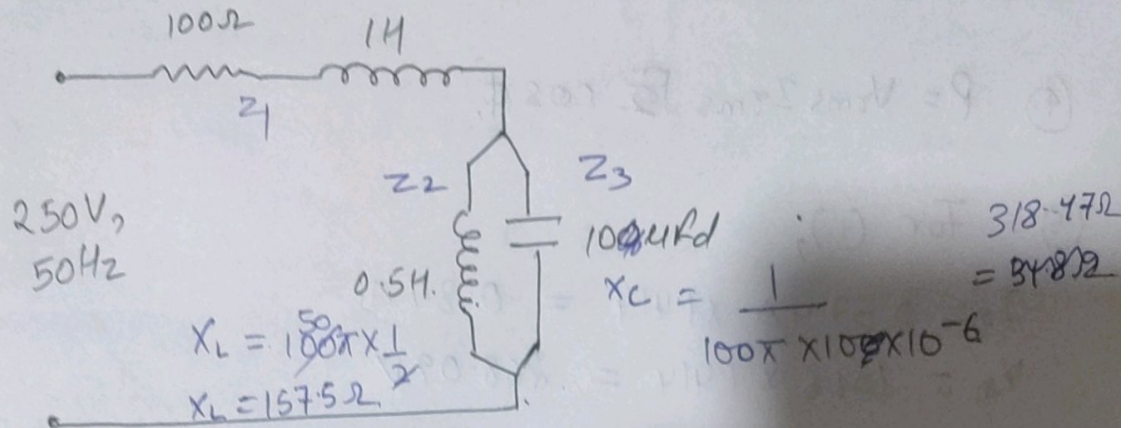
$$(I_{rms})_T = 14.2A$$

$$Z_T = 13.9$$

$$\boxed{Z_T = 14\Omega}$$

Q2. With reference to network given below find.

- ① Z_T
- ② Branch Current
- ③ Power Factor
- ④ Power dissipated
- ⑤ Phasor.



Ans $Z_1 = \sqrt{R^2 + X_L^2}$

$$Z_1 = \sqrt{10000 + (314.15)^2}$$

$$Z_1 = 330.01\Omega$$

$$Z_2 = 157.08\Omega$$

$$Z_3 = 318.47\Omega$$

$$Z_T = 330.01 + \frac{157.08 \times 318.47}{318.47 + 157.08} = \frac{50004.5}{475.55} + 330.01$$

$$Z_T = 435.06\Omega$$

② $I_T = \frac{250}{435.06}$

$$I_T = 0.574A = I_1$$

$$I_1 = \frac{V_1}{Z_1} \Rightarrow V_1 = 0.574 \times 330.01$$

$$V_1 = 189.6V$$

$$V_2 = V_3 = 60.3V$$

$$I_2 = \frac{60.3}{157.08} = 0.38$$

$$I_3 = \frac{60.3}{318.47} = 0.189$$

$$(3) \cos \phi = \frac{R_T}{Z_T} \quad R_T = 100 \Omega$$

$$\cos \phi = \frac{100}{435.06} =$$

$$\frac{60}{16} + 4$$

$$\frac{124}{16.84}$$

$$(4) P = V_{rms} I_{rms} \cos \phi$$

(5) For (1);

$$I_m = 0.574 A \times 1.414 = 0.811 A$$

$$V_m = 189.6 \times 1.414 = 268.09 V$$

$$\cos \phi_1 = \frac{R_1}{Z_1} = \frac{100}{330.01 \Omega} = 0.303$$

$$\boxed{\phi_1 = 72.3^\circ}$$

For (2)

$$(I_m)_2 = 0.38 \times 1.414 = 0.537 A$$

$$(V_m)_2 = 60.3 \times 1.414 = 85.26 V$$

$$\cos \phi_2 = 0$$

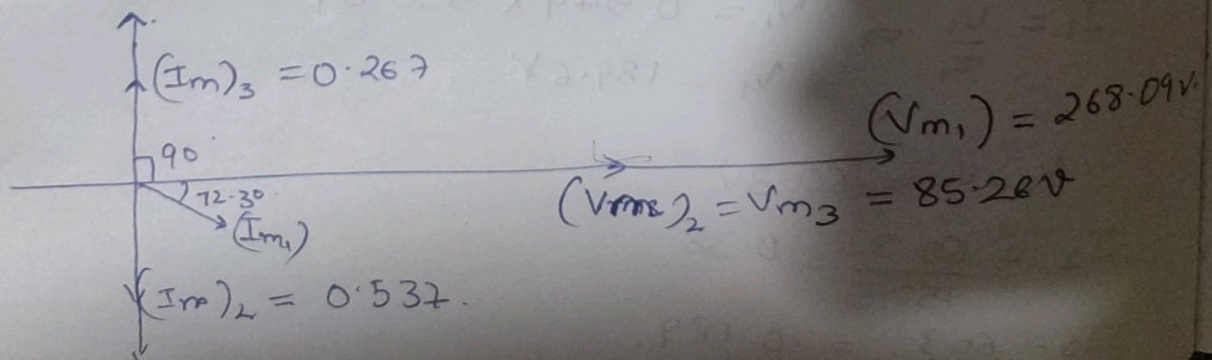
$$\boxed{\phi_2 = 90^\circ = \phi_3}$$

For (3)

$$(I_m)_3 = 0.189 \times 1.414 = 0.267 A$$

$$(V_m)_3 = 85.26 V$$

Phasor

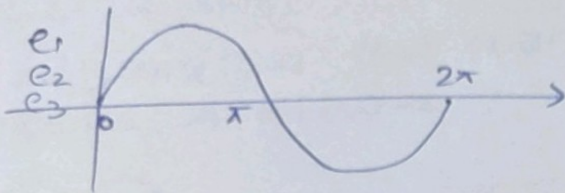


* THREE PHASE AC SYSTEM

• Generation of three phase AC: -

→ Whenever there is relative motion b/w set of conductors and uniform magnetic field. then emf is generated and that emf is of three phase AC.

① When ϕ Phase diffⁿ b/w their emf is zero then



$$e_1 = e_2 = e_3$$
$$N_1 = N_2 = N_3$$

But for continuous supply of electricity, we need uninterrupted three phase AC. So we have to take phase difference b/w their emf.

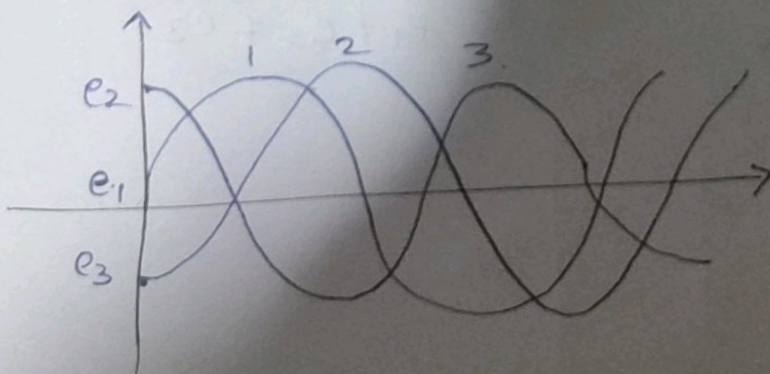
$$\text{Phase difference} = \frac{360 \text{ electrical degrees}}{\text{Angle} \quad N (\text{No of phase})}$$

$$\phi = \frac{360}{N}$$

If three phase are rep^d.

$\phi = 120^\circ$ → means for uniformity we need ϕ difference of 120° with each other.

* Waveform for 3 phase AC: -



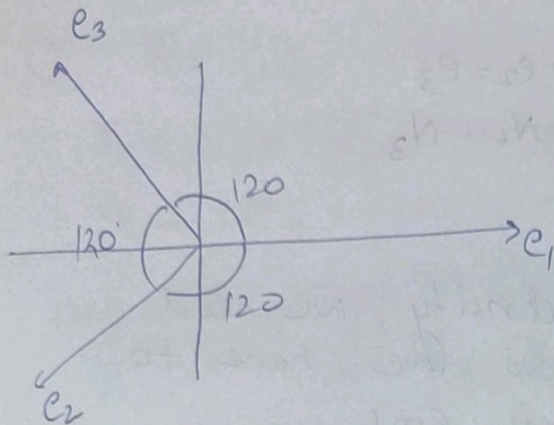
Now we get continuous power supply
↓
Uninterrupted.

$$e_1 = E_m \sin \omega t$$

$$e_2 = E_m \sin \left(\omega t - \frac{2\pi}{3} \right)$$

$$e_3 = E_m \sin \left(\omega t - \frac{4\pi}{3} \right)$$

* Phasor of e_1, e_2, e_3



* Phase Sequence:-

It is a sequence or order in which alternating quantities attain their positive peak values.

Sequence: R Y B
(Red) (Yellow) (Blue)

* Types of 3 phase AC:-

(Balanced).
1. Symmetrical AC system

- No of turns of all conductor are same.
- dirⁿ of rotation is same
- Amt of flux cut by each conductor set is same.

↓

"Emf produced by each supply conductor are same".

1. Magnitude of emf is same
2. ϕ angle b/w each conductor is 120°

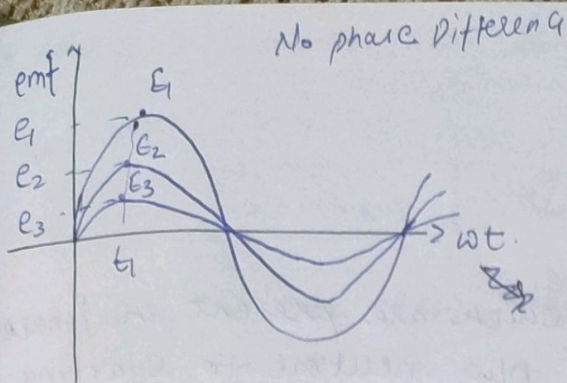
(Unbalanced).
2. Unsymmetrical AC system

- No of turns of each conductor is different

↓

- Magnitude of emf due to each conductor is different.

$$e_1 \neq e_2 \neq e_3$$

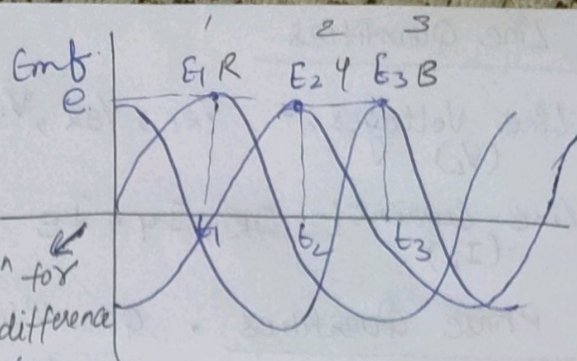


→ time taken to attain peak value is same.

but difference is that their magnitude is different

R. Y B → +ve phase Sequence

B. Y R → -ve phase Sequence



Condⁿ for phase difference

$$E_m = E_{m1} = E_{m2} = E_{m3}$$

The difference is that time taken to attain peak value is different

$$t_1 \neq t_2 \neq t_3$$

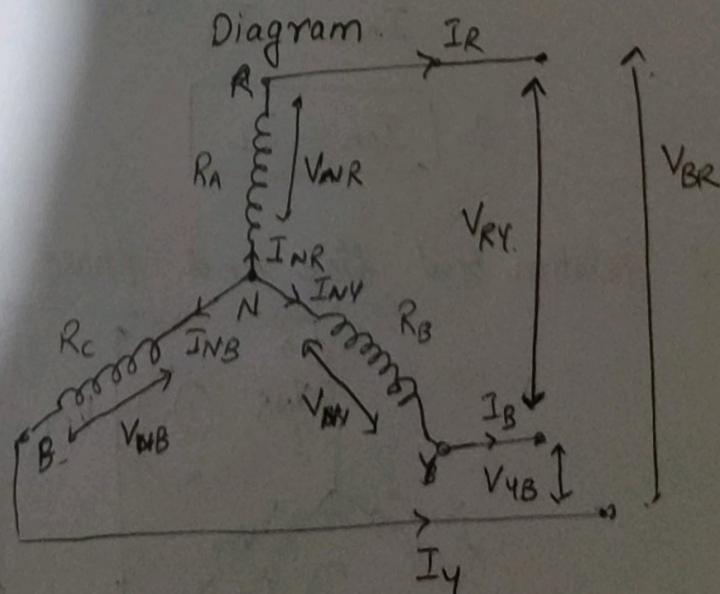
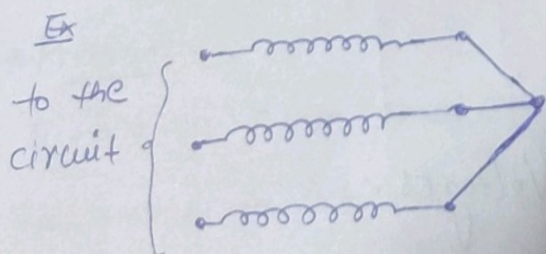
but magnitude is same.

Connection of 3 phase AC

1. Star Connection
2. Delta Connection.

1. Star Connection: (For Balanced Circuit).

It is a type of connection where one terminal from each winding is connected to a common point and remaining three terminal are connected to the circuit.



* Line Quantities

1. Line Voltages:- V_{RY}, V_{BR}, V_{YB}
(V_L)
2. Line Current:- I_R, I_Y, I_B
(I_L)

* Phase Quantities : Quantities which are present in inner section or b/w neutral to starting

1. Phase Voltages:- $V_{NR}, V_{NY}, V_{NB} = V_{ph}$.
2. Phase current:- $I_{NR}, I_{NY}, I_{NB} = I_{ph}$.

For Balanced Circuits,

1. Line voltages: $V_{RY} = V_{BR} = V_{YB} = V_L$.
2. Line Current: $I_R = I_Y = I_B = I_L$.
3. Phase voltages: $V_{NR} = V_{NY} = V_{NB} = V_{ph}$.
4. Phase current: $I_{NR} = I_{NY} = I_{NB} = I_{ph}$.

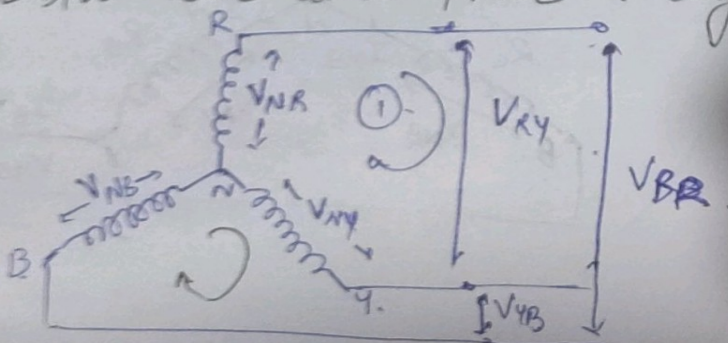
* Relation b/w line and phase quantities for balanced 3 ϕ star connected system.

1. Relation b/w line ^{current} voltages & phase ^{current} voltages.

~~For~~
$$\begin{aligned} I_{NR} &= I_R \\ I_{NY} &= I_Y \\ I_{NB} &= I_B \end{aligned} \quad \left\{ \begin{array}{l} \text{by series} \\ \text{connection} \end{array} \right\}$$

So
$$I_{ph} = I_L$$

2. Relation b/w line and Phase Voltages.



• Applying KVL in Mesh NRYN

$$V_{NR} + V_{RY} - V_{NY} = 0$$

$$\boxed{\vec{V}_{RY} = \vec{V}_{NY} - \vec{V}_{NR}} \quad \text{--- (1)}$$

• Applying KVL in Mesh NYBN

$$+V_{NY} + V_{YB} - V_{NB} = 0$$

$$\boxed{\vec{V}_{YB} = \vec{V}_{NB} - \vec{V}_{NY}} \quad \text{--- (2)}$$

• Applying KVL in Mesh BNRBN

$$V_{NB} - V_{BR} - V_{NR} = 0$$

$$\boxed{\vec{V}_{BR} = \vec{V}_{NR} - \vec{V}_{NB}} \quad \text{--- (3)}$$

From eqⁿ (1),

$$V_{RY} = \sqrt{V_{NY}^2 + V_{NR}^2 + 2V_{NY}V_{NR}\cos\phi}$$

$$V_L = V_{RY} = \sqrt{2V_{ph}^2 + 2V_{ph}^2(\cos\phi)}$$

$$V_L = \sqrt{2V_{ph}^2(1+\cos\phi)}$$

$$V_L = \sqrt{2V_{ph}^2(1+\cos 60^\circ)}$$

$$V_L = \sqrt{2V_{ph}^2(1+\frac{1}{2})} = \sqrt{3}V_{ph}$$

$$\boxed{V_L = \sqrt{3}V_{ph}}$$

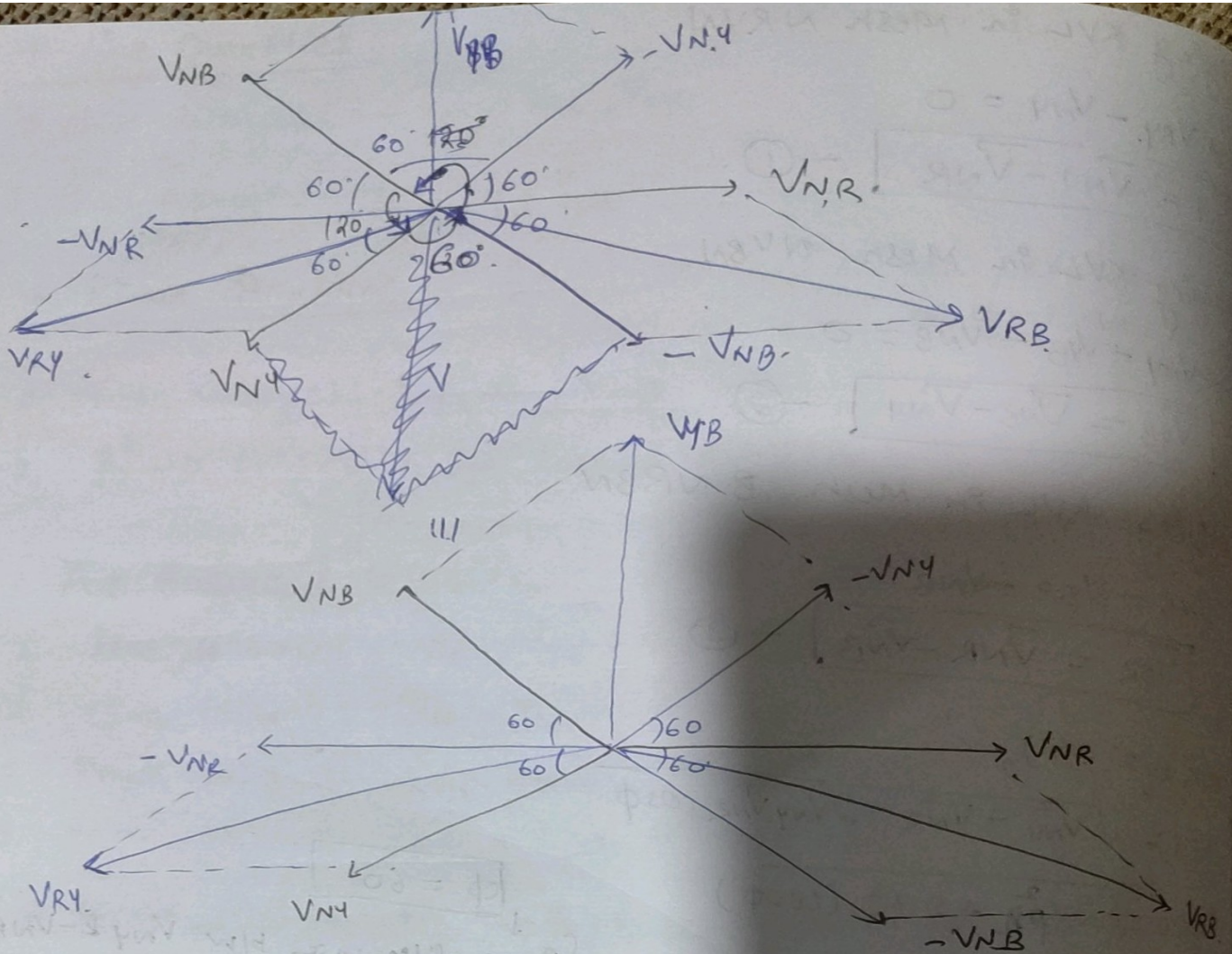
* Phasor

R → Y → B (+ve phase sequence).

$V_{NR} \rightarrow V_{NY} \rightarrow V_{NB}$

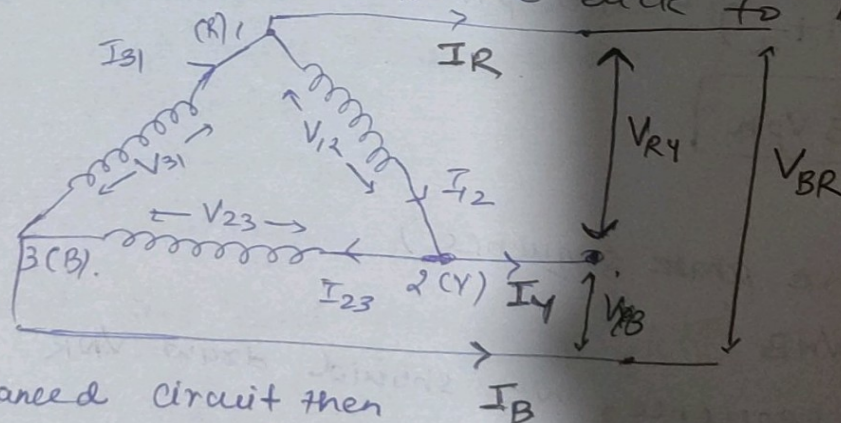
Acc to phase sequence, we should draw V_{NR} .

$\phi = 60^\circ$
↓
Phase difference b/w V_{NY} & V_{NR}



2. DELTA CONNECTION (Δ)

→ It is a type of connection of 3 phase winding where the coils are connected in a back to back arrangement.



As it is balanced circuit then

- (a) Line voltages: $V_{RY} = V_{BR} = V_{YB} = V_L$
- (b) Line current: $I_R = I_Y = I_B = I_L$
- (c) Phase voltages: $V_{12} = V_{23} = V_{31} = V_{ph}$
- (d) Phase current: $I_{12} = I_{23} = I_{31} = I_{ph}$

(a) Relation b/w line & phase Voltages

$$V_L = V_{RY} = V_{YB} = V_{BR}$$

$$V_{ph} = V_{12} = V_{23} = V_{31}$$

$$\Rightarrow \boxed{\begin{matrix} V_{RY} = V_{12} \\ V_{YB} = V_{23} \\ V_{BR} = V_{31} \end{matrix}} \quad \left\{ \text{by observation} \right\}$$

$$\Rightarrow \boxed{V_L = V_{ph}}$$

(b) Relation b/w line and phase Currents

$$I_L = I_R = I_Y = I_B$$

$$I_{ph} = I_{12} = I_{23} = I_{31}$$

By using KCL, we get relation b/w I_L & I_{ph} .

① By KCL at $j^n R$.

$$\text{Incoming current} = \text{Outgoing current}$$

$$\Rightarrow I_{31} = I_R + I_{12}$$
$$\boxed{\vec{I}_R = \vec{I}_{31} - \vec{I}_{12}} \quad \text{--- (1)}$$

② KCL at $j^n Y$.

$$I_{12} = I_Y + I_{23}$$

$$\boxed{\vec{I}_Y = \vec{I}_{12} - \vec{I}_{23}} \quad \text{--- (2)}$$

③ KCL at $j^n B$.

$$I_{23} = I_{31} + I_B$$

$$\boxed{\vec{I}_B = \vec{I}_{23} - \vec{I}_{31}} \quad \text{--- (3)}$$

Taking eqn (i)

$$\bar{I}_R = \bar{I}_{31} - \bar{I}_{12}$$

$$I_R = \sqrt{I_{31}^2 + I_{12}^2 + 2I_{31}I_{12}\cos\phi}$$

$$I_L = \sqrt{2I_{ph}^2 + 2I_{ph}^2\cos\phi}$$

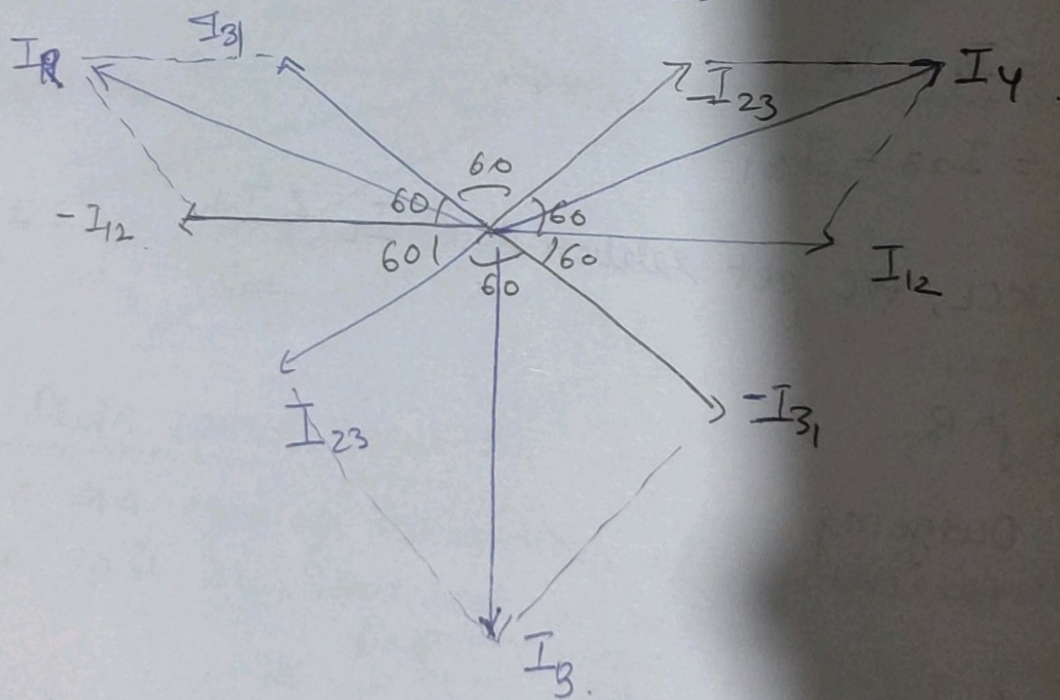
$$I_L = I_{ph} \sqrt{2(1+\cos\phi)}$$

$$I_L = \sqrt{3} I_{ph}$$

* Phasor.

Acc to phase sequence

$$I_{12} \rightarrow I_{23} \rightarrow I_{31}$$

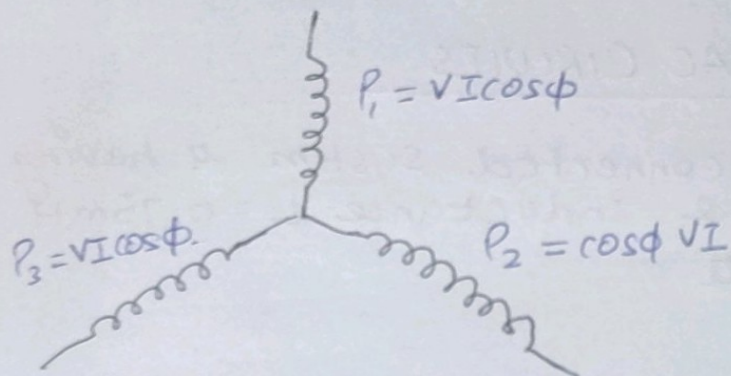


POWER IN 3 ϕ AC

(a) Active Power (P) = $V_{rms} I_{rms} \cos \phi$

(b) Reactive Power (Q) = $V_{rms} I_{rms} \sin \phi$

(c) Apparent Power (S) = $V_{rms} \cdot I_{rms}$.



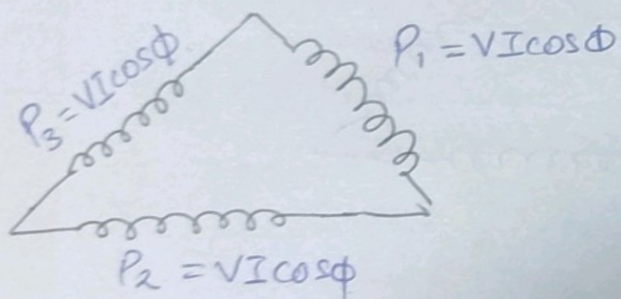
$$P_T = 3 \frac{V_L}{\sqrt{3}} \cdot I_L \cos \phi$$

$$P_T = \sqrt{3} V_L I_L \cos \phi$$

$$V_1 = V_2 = V_3 = V_{ph}$$

$$I_1 = I_2 = I_3 = I_{ph}$$

$$P_T = 3 V_{ph} I_{ph} \cos \phi \quad (\text{star connection}).$$



$$\therefore V_1 = V_2 = V_3 = V_{ph}$$

$$I_1 = I_2 = I_3 = I_{ph}$$

$$P_T = 3 V_{ph} I_{ph} \cos \phi \quad (\text{Delta connection}).$$

$$P_T = 3 V_L \cdot \frac{I_L}{\sqrt{3}} \cos \phi$$

$$P_T = \sqrt{3} V_L I_L \cos \phi$$

Active Power - $P = 3V_{ph}I_{ph}\cos\phi$

Reactive Power - $Q = 3V_{ph}I_{ph}\sin\phi$

Apparent Power - $S = 3V_{ph}I_{ph}$

$$P = \sqrt{3}V_L I_L \cos\phi$$

$$Q = \sqrt{3}V_L I_L \sin\phi$$

$$S = \sqrt{3}V_L I_L$$

NUMERICALS ON 3 PHASE AC CIRCUITS

Q1. If a 3 ϕ Balanced star connected system is having a coil of Resistance 10Ω & Inductance $L = 0.75mH$ at supply of 3A. Calculate

(a) Total Impedance.

(b) Total Supply Voltage.

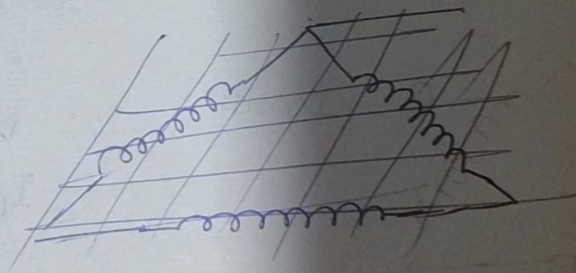
(c) Net Power.

(d) Phasor

Ans $R = 10\Omega$, $L = 0.75mH$, $I_L = 3A$.

(1) ~~$Z = \sqrt{R^2 + X_L^2}$~~

~~$X_L = \omega L$~~



For star connection,

$$V_L = \sqrt{3}V_{ph}$$

$$I_L = I_{ph}, \quad I_L = 3A \text{ \{Given\}}$$

* Note:-

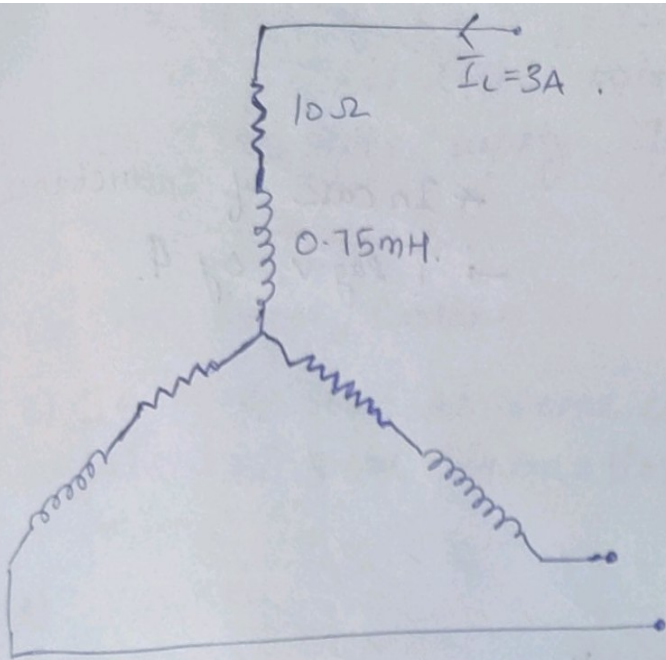
→ Impedance is always given in terms of phase.

Z_{phase}

With help of Impedance, we find current by formula

$$I_{ph} = \frac{V_{ph}}{Z_{ph}}$$

then I_{ph} , V_{ph} are used



Given;

$$R = 10\Omega$$

$$L = 0.75 \times 10^{-3} \text{ H}$$

$$X_L = \omega L$$

$$X_L = 2\pi f L$$

* Note:-

If frequency is not given, then by default, we assume

$$f = 50 \text{ Hz}$$

$$X_L = 2 \times 3.14 \times 50 \times 0.75 \times 10^{-3}$$

$$X_L = 0.2355\Omega$$

$$(a) Z = \sqrt{R^2 + X_L^2}$$

$$Z_{ph} = \sqrt{100 + (0.235)^2} = \sqrt{100 + 0.055}$$

$$Z_{ph} = \sqrt{100.055} = 10.002\Omega$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}}$$

$$I_{ph} = I_L = 3 \text{ A [Star conn]}]$$

$$3 \times 10.002 = V_{ph}$$

$$V_{ph} = 30.008 \text{ V}$$

$$\text{OR } V_L = \sqrt{3} \times 30.008$$

$$V_L = 51.9 \text{ V}$$

Total supply voltage

$$(b) \text{ Net power} = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} \times 51.9 \times 3 \times \cos \phi$$

$$\cos \phi = \frac{R}{Z} = \frac{10}{10.002} = 0.9998$$

ϕ

$$P = 1.73 \times 51.9 \times 3 \times 0.999$$

$$P = 269.30 \text{ watt}$$

(d). Phasor

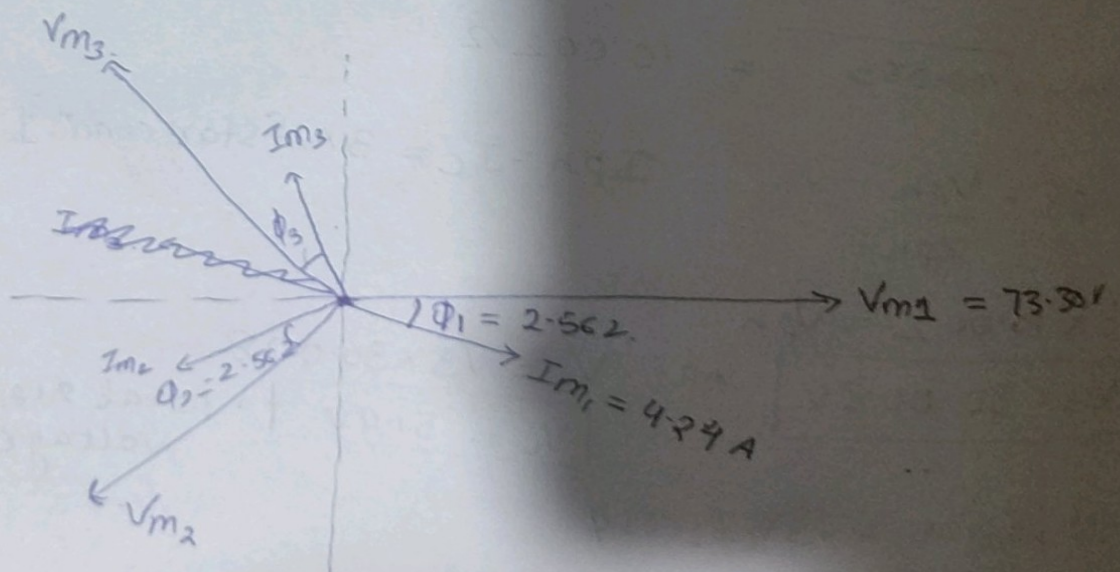
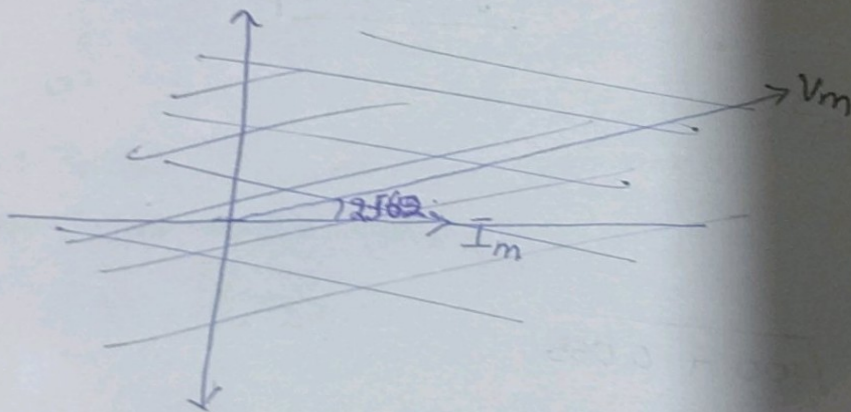
$$V_m = V_{rms} \cdot \sqrt{2} = V_L \cdot \sqrt{2} \\ = 51.9 \times 1.414 = 73.38$$

$$I_m = I_{rms} \cdot \sqrt{2} = I_L \cdot \sqrt{2} \\ = 3 \times 1.414 = 4.242$$

$$\phi = \cos^{-1}(0.9998)$$

$$\phi = 2.562^\circ$$

* In case of Inductance
→ i lag v by ϕ .



2. If a load of $z = 3 - j7 \Omega$ in series with an inductor of 5Ω is connected in balanced 3 phase delta conn across $440V, 50Hz$ supply. Then calculate

(a) Total Impedence.

(b) Net Supply Current

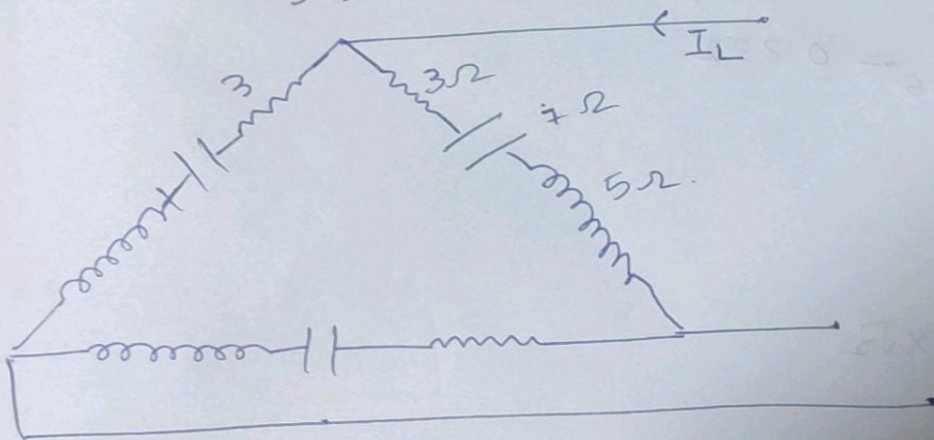
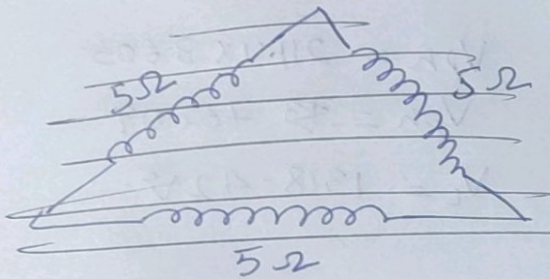
(c) If same load at same current is connected in 3 phase balanced star formation then find its KVAR rating.

(d) Phasor.

* Note:-

Load consist of Resistance and Capacitor or Inductor (Choose this acc to leading or lagging sign)

Ans



$$\textcircled{1} Z_T = \sqrt{R^2 + (X_C - X_L)^2}$$

$$Z_{ph} = \sqrt{9 + (7 - 5)^2}$$

$$= \sqrt{9 + 4 + 0}$$

$$= \sqrt{13}$$

$$= 3.605$$

$$\textcircled{2} V_L = 440V$$

$$I_L = ?$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}}$$

$$\phi I_{ph} = \frac{440}{3.605} = 122.05A$$

$$I_L = \sqrt{3} I_{ph}$$

$$I_L = 211.1A$$

(3) Now in star connection,
 ~~$V_{ph} = \frac{V_L}{\sqrt{3}}$~~ $V_L = \sqrt{3} V_{ph}$
 $I_L = I_{ph}$

$$\frac{I}{I_{ph}} = \frac{V_{ph}}{Z_{ph}}$$

$$I_{ph} = \frac{231.2}{3.605} = 64.14$$

$$Q = \sqrt{3} V_{rms} \cdot I_{rms} \cdot \sin \phi$$

$$Q = \sqrt{3} V_L I_L \sin \phi$$

$$Q = 400 \times 64.14 \times \sqrt{3}$$

$$\cos \phi = \frac{R}{Z} = \frac{3}{3.605} = 0.832$$

$$\sin \phi = \sqrt{1 - \cos^2 \phi}$$

$$\sin \phi = \sqrt{1 - 0.692}$$

$$Q = \sqrt{3} V_L \cdot I_L \cdot \sin \phi$$

$$Z_{ph} = 3.605$$

$$I_L = 211.4 \text{ A}$$

$$I_{ph} = 211.4 \text{ A}$$

$$V_{ph} = 211.4 \times 3.605$$

$$V_{ph} = 762.09$$

$$V_L = 1318.42 \text{ V}$$

$$Q = 1.73 \times 1318.42 \times 211.4 \times \sin \phi$$

$$\cos \phi = \frac{R}{Z} = \frac{3}{3.605} = 0.832$$

~~Answer~~

(4) Phasor

$$V_m = V_{rms} \times \sqrt{2} = V_L \times \sqrt{2}$$

$$V_m = 622.16$$

$$I_m = I_L \times \sqrt{2}$$

$$= 211.4 \times \sqrt{2} = 298.9$$

$$I_m = 298.9 \text{ A}$$

$$\phi = 33.695^\circ$$

Phasor diagram

